Cell Cryopreservation

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SUSTech-BIO 2026 iGEM Team

Purpose

Prepare frozen stocks of healthy mammalian cells using Servicebio G1711 serum/protein-free, DMSO-free non-gradient cell freezing medium. Our stocks are frozen and stored at −80°C. Density selection covers HEK293T, U251, SH-SY5Y and CCF-STTG1.

Materials & Equipment

Item	Specification / use
Serum/protein-free non-gradient cell freezing medium, DMSO-free	Servicebio G1711; ready to use. Use the supplied formulation without adding serum or DMSO. [6]
Healthy cells; matching complete medium	Use the cell-specific maintenance conditions in C01.
PBS/DPBS; dissociation reagent	Ca2+/Mg2+-free wash buffer and the established dissociation reagent for adherent cells.
Cell counter and viability stain	Determine viable cells, not only total particles.
Cryogenic vials; sterile tubes and pipettes	Sterile cryovials rated for −80°C storage; 1 mL suspension per vial is the working example used here.
Freezer-compatible vial holder; optional controlled-rate container	Use a holder for direct −80°C freezing. A controlled-rate container is not required for this medium; discussed below for comparison.
Culture cabinet; centrifuge; microscope	Eppendorf refrigerated centrifuge; aseptic collection and inspection.
−80°C freezer	Cell freezing and storage; maintain a stable temperature.
Pipette tips	Axygen; sterile tips compatible with the selected pipettes.

Before You Start

Check the medium and culture

Use the Servicebio G1711 medium as supplied; gently mix before use and do not add serum, DMSO or basal medium. Store unused reagent at 2–8°C, protected from light; the stated unopened shelf life is 12 months, with use within 3 months after opening. The manufacturer also permits aliquoted reagent storage at −20°C. These are reagent-storage conditions, not a staged cell-freezing programme. [6]

Use actively growing, uncontaminated cultures with normal morphology. Check identity, passage and mycoplasma status. Avoid banking stressed or overgrown cultures merely to rescue them. Label vials and reserve a storage position before adding cryoprotectant. [1]

Choose a density: reference ranges, not proven optima

Servicebio specifies 0.8–5 × 10^6 cells/mL for G1711. For healthy cultures, use viable-cell counts to choose a working density within this range. The four adherent lines below retain 2–5 × 10^6 viable cells/mL as a practical starting range, within the product range and consistent with general ATCC guidance. These are evaluation ranges, not experimentally established optima for each line. [1, 6]

Cell type	Reference density	How to apply it
HEK293T	2–5 × 10^6 viable cells/mL	General mammalian monolayer starting range within the Servicebio product range. Assess post-thaw recovery. [1, 6]
U251	2–5 × 10^6 viable cells/mL	Same general starting range within the Servicebio product range; no distinct line-specific optimum is established here. [1, 6]
SH-SY5Y, undifferentiated	2–5 × 10^6 viable cells/mL	General starting range. A supplier uses about 3 × 10^6 cells/mL for a GFP derivative in another medium; this is supporting context, not validation for parental cells or Servicebio. [1, 3]
CCF-STTG1	2–5 × 10^6 viable cells/mL	General starting range. Its retrieved supplier sheet gives a freezing formulation but no freezing density. Check recovery before banking the full stock. [1, 7]
PBMCs (comparison only)	0.5–10 × 10^6 cells/mL	For CryoStor CS10 or 90% FBS/10% DMSO only. Do not transfer this upper limit to Servicebio, whose stated range ends at 5 × 10^6 cells/mL. [4, 6]
Other primary cells, stem cells or differentiated neurons	No common range assigned	Use a matched cell/formulation protocol. Single cells and aggregates require different counting and recovery criteria. Do not transfer the tumour-cell ranges without validation.

For a first comparison, vary density within the applicable range while holding medium, vial fill, cooling and thawing constant. Select the condition by viable cell recovery and subsequent attachment/growth or relevant function, not immediate viability percentage alone. A high survival percentage can conceal a low total recovery.

Decide whether to use a freezing container

Situation	Choice and trade-off
Servicebio G1711: routine stock	A controlled-rate freezing container is not required by the Servicebio instructions. Place aliquoted vials directly at −80 °C. Our stocks remain in the −80 °C freezer. [6]
Considering a container with Servicebio	Adding a container changes the cooling profile. The instructions do not establish a recovery advantage; use it only as a deliberately evaluated alternative, with density and fill held constant.
Conventional slow-cooling formulation	Use an appropriate container or controlled-rate freezer, typically around −1 °C/min. This comparison is not an extra stage in the Servicebio procedure. [2, 5]
New, sensitive or valuable stock	Keep a recoverable backup while assessing a representative vial. A different cell-specific formulation or cooling protocol may be needed if direct freezing gives poor recovery.

For the optional slow-cooling comparison, a freezing container is a passive device, not an electronically programmed freezer. An ordinary cryobox does not provide a controlled cooling rate. Mr. Frosty specifically starts at room temperature, uses 100% isopropanol and requires alcohol replacement after five uses; alcohol-free devices have their own instructions. These preparations are unnecessary for the main direct-freezing route. [5]

Procedure

A. Collect and count

1. Harvest adherent cells using C01: wash, detach gently, neutralize the dissociation reagent and collect in complete medium. For a suspension line, collect the suspension directly using its established harvesting procedure.
2. Mix gently, take a representative sample and measure viable cell concentration and total suspension volume. Record cell identity, passage and pre-freeze viability. Calculate the total viable cell number before choosing the number of vials.
3. Pellet cells gently and remove the supernatant without disturbing the pellet. The Servicebio instructions state 800–1200 rpm for 3–5 min; rpm depends on rotor radius and is not interchangeable with × g. The C01 condition of 200 × g for 5 min may be retained as a laboratory adaptation where effective for the cell line, rather than presented as a conversion of the supplier's setting. [2, 6]

B. Resuspend and aliquot

4. Calculate the final freezing-suspension volume: V (mL) = total viable cells / target viable cells per mL. For example, 1.2 × 10^7 viable cells at 3 × 10^6/mL require 4 mL final suspension, giving four 1 mL aliquots. This is arithmetic, not a confirmed team density.
5. Gently mix the ready-to-use G1711 medium, then resuspend the pellet in the calculated volume. Proceed promptly; no additional warming or equilibration hold is specified. [6]

6. Dispense 1 mL per compatible cryovial, as recommended by Servicebio. Mix gently between aliquots to limit settling; record cells/mL and cells/vial. Close vials securely and proceed promptly to freezing. [6]

C. Freeze and store at −80°C

7. Place the cryovials directly in a stable −80°C freezer using a suitable vial holder. Do not use a controlled-rate freezing container for this standard route, and do not add 4°C or −20°C holding stages. Keep the freezer temperature stable and minimize door opening. [6]

8. Keep the frozen vials in the −80°C freezer. Record the freezing date and box/vial position, and record storage duration when a vial is retrieved for recovery. Minimize unnecessary handling and temperature fluctuations.

9. Recover a representative vial using C03 for compatible routine adherent lines. Dilute the thawed suspension in complete medium before centrifugation: the Servicebio medium is viscous and should not be centrifuged undiluted. Assess viable recovery, attachment and subsequent growth/morphology. Keep the original culture or backup stock until the new batch has passed recovery assessment. [6]

p.s.

Prepare labels, the vial holder and freezer space before resuspending cells in freezing medium. Avoid leaving the final tubes waiting while setting up the next step.

Use G1711 as supplied. It is DMSO-free; do not add DMSO, serum or basal medium to this ready-to-use reagent. Gently mix the reagent before resuspending the cells. [6]

Higher density is not automatically better: clumps impair counting and even aliquoting. A sparse vial may also yield too few recoverable cells. Compare recovery under the same conditions before narrowing the range.

For bank qualification, record the G1711 lot, viable cells per vial, fill volume, cooling route and thawing observations. Avoid changing density and cooling route simultaneously when diagnosing a poor batch.

Expected Outcome

A qualified stock should recover viable cells that resume the expected morphology and growth after thawing. Report viable recovery as recovered viable cells divided by the viable cells originally frozen, using a defined counting time; assess later attachment and expansion separately. No fixed recovery percentage or Servicebio-specific optimum is claimed.

Troubleshooting

Observed problem	Check first	Action
Clumps or unequal cells per vial	Dissociation quality, counting and settling during dispensing	Use gentle dispersion and regular mixing; recount a representative suspension. Do not force damaged aggregates through repeated harsh pipetting.
Low viable recovery after thawing	Pre-freeze health, medium identity, actual cooling route, storage and C03 thawing records	Correct the identifiable fault first. Compare density within the supported range with the other conditions fixed.
Good immediate viability but poor attachment or growth	Recovered cell number, reseeding area, handling damage and appropriate recovery medium	Assess viable yield and next-day culture as well as dye exclusion. Use a cell-appropriate recovery density and vessel.
One batch performs worse than earlier batches	Medium lot, vial fill, freezer temperature, unintended use of an insulating container and warming during handling	Restore the established direct-freezing setup. If comparing a container, keep other conditions fixed and assess recovery before adopting the change.
Cracked, leaking or poorly sealed vial	Cryogenic rating, fill limit and closure	Exclude compromised vials from the bank and review compatible vial handling. Use the laboratory cryogenic incident procedure.

Reference

[1] ATCC. Cryogenic Storage of Animal Cells: culture health and general 2–5 million viable cells/mL reference range. Accessed 2026-10-03. Open source

[2] Thermo Fisher Scientific / Gibco. Cell Freezing Protocols: collection and conventional slow-cooling comparison. Accessed 2026-10-03. Open source

[3] GenTarget. SH-SY5Y/GFP stable cell line SC042 manual, copyright 2024, p. 2: approximately 3 × 10^6 cells/mL in 90% complete medium/10% DMSO. Derivative-line context only. Accessed 2026-10-03. Open source

[4] STEMCELL Technologies. Protocol for Cryopreserving PBMCs, PR00022 v1.0.1, June 2024: 0.5–10 × 10^6 cells/mL and recovery evaluation. Accessed 2026-10-03. Open source

[5] Thermo Scientific. Mr. Frosty Freezing Container, 5100-0001 product page and FAQs: cooling rate, room-temperature setup and isopropanol replacement. Accessed 2026-10-03. Open source

[6] Servicebio. G1711 Serum/Protein-Free Non-gradient Cell Freezing Medium (without DMSO), V1.1-20260526, Chinese pp. 1–2 and English pp. 1–2. Official IFU accessed 2026-10-03. Open source

[7] Procell. CCF-STTG1 cell instructions, CL-1022, Rev. V1.0, p. 1: freezing formulation specified without a freezing-density recommendation. Accessed and downloaded 2026-10-03. Open source